BASICS OF FIM · WI001287

Statistical Learning

4: Logistic regression & Bayes' classification

Univ.-Prof. Dr.-Ing. habil. Niklas Kühl
Lars Böcking

Chair for Information Systems and
Human-centric Artificial Intelligence
University of Bayreuth
TUM School of Management

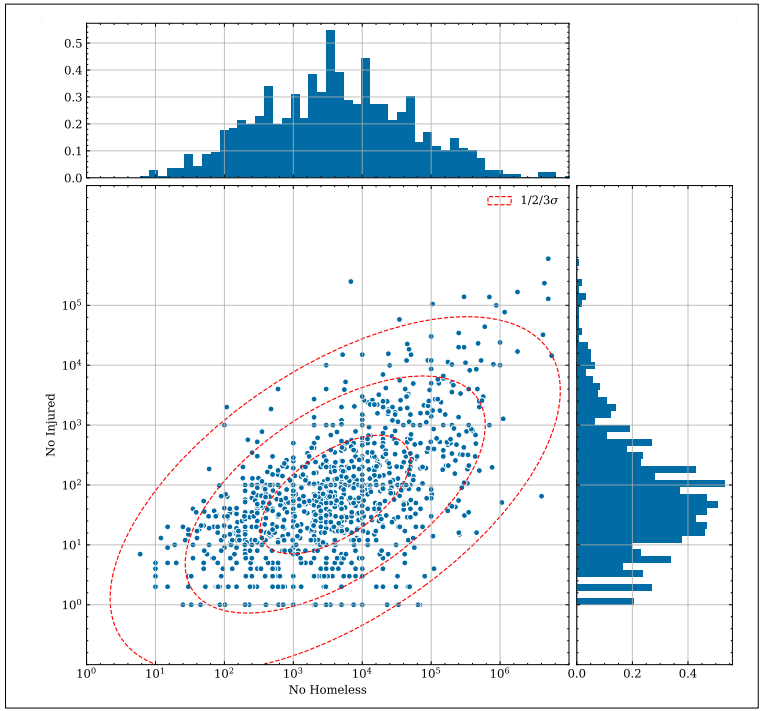


Figure 4.5: multivariate

Agenda.

- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

Agenda.

- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

Recap: Lecture 2 Regression vs. Classification

Measurable Machine Learning Problems

Business Problem

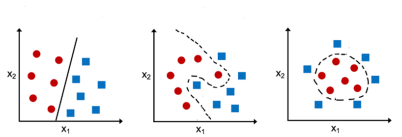
Objectives and business **success criteria** describe the problem. However, these descriptions are often subjective and lack measurability.

Translation

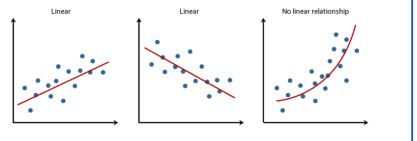
Machine Learning Problem

Machine learning algorithms on the other hand require a **clear, measurable goal** that can be quantified by a dominate metric.

Classification | *classify instances into classes*



Regression | *predict numerics as close as possible*



19

Statistical Learning | Prof. Dr. Niklas Kühl

Recap: Lecture 3 introduced linear regression

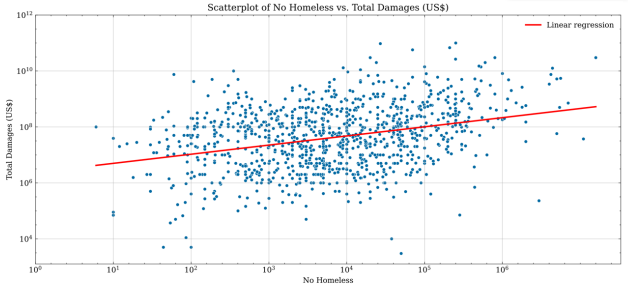
We bet the world is a line plus noise.

Linear regression is a simple approach to supervised learning. We believe about the world that: $Y \approx f(X) + \varepsilon$. In the case of linear regression, we assumes that the dependence of Y on X is linear:

y-axis intercept *error term*

$$Y = \beta_0 + \beta_1 \times X + \varepsilon$$

slope



… but we can not observe this entity
… instead we have a finite set of observations $\{(x_i, y_i)\}_{i=1}^{N}$
… from which we learn the function:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$$

True regression functions are never linear!

Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

What if we want to predict the type of disaster?

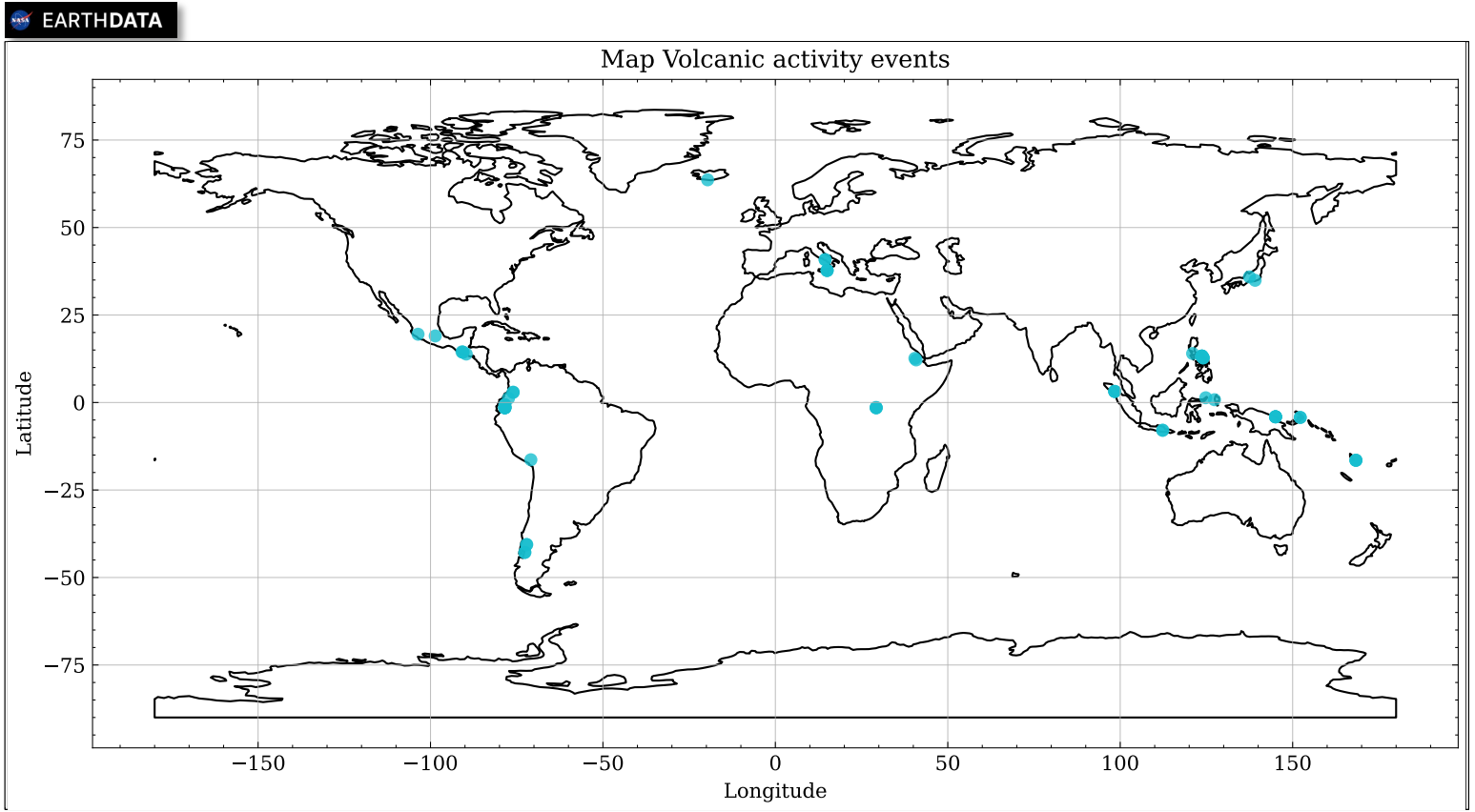
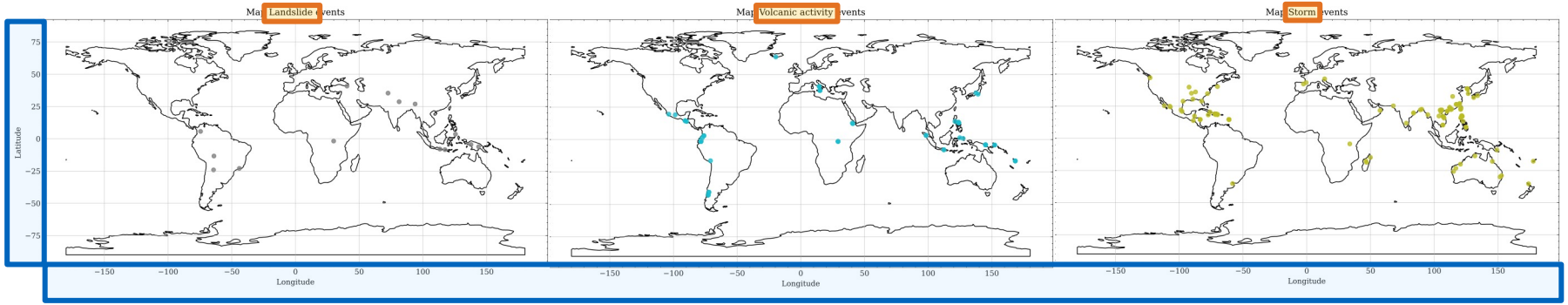


Figure 4.1: world map

What if we want to predict the type of disaster?



Feature: an individual measurable property or characteristic of a phenomenon

Target: is the variable the model is trying to predict

Can we use linear regression to predict disaster types?

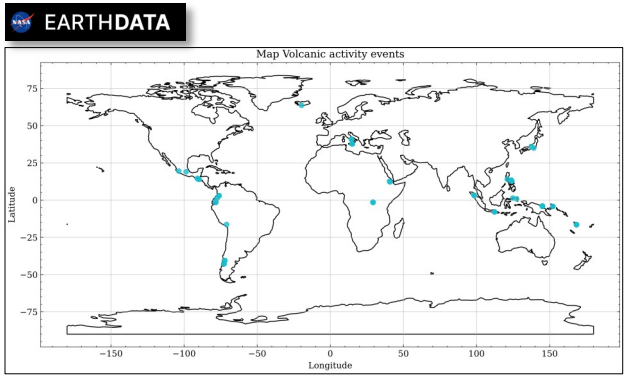


Figure 4.1: world map

Can we simply perform a **linear regression** of Y on X and classify as volcanic activity if $\hat{Y} > 0.5$?

Does the linear model fit our task?

Let's define our classification task as:

$$Y = \begin{cases} 1 \; if \; volcanic \; activity \\ 0 \; else \end{cases}$$

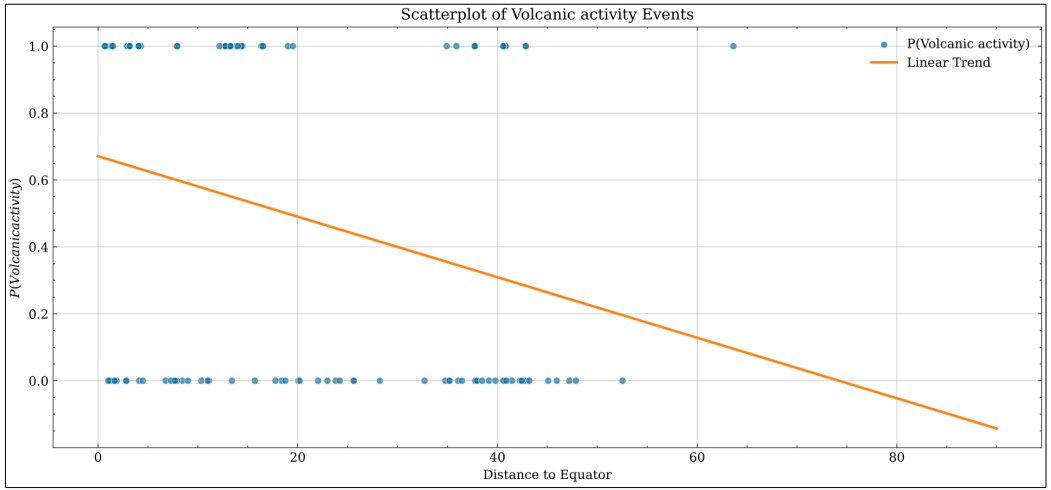


Figure 4.2: linear attempt

Can we use linear regression to predict disaster types?

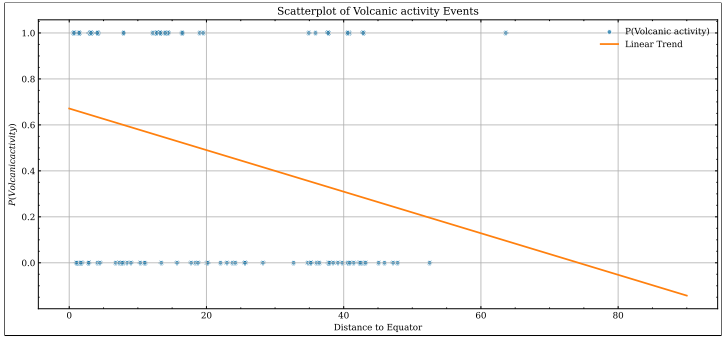
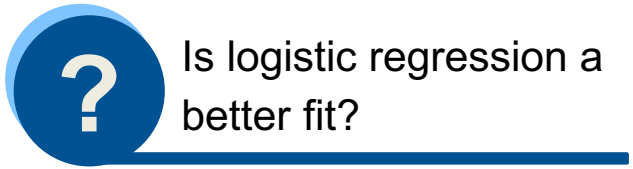


Figure 4.2: linear attempt

We want to ensure that: $P(X) \in [0,1]$

→ We need a link function that relates the linear model to the parameter of the outcome distribution.



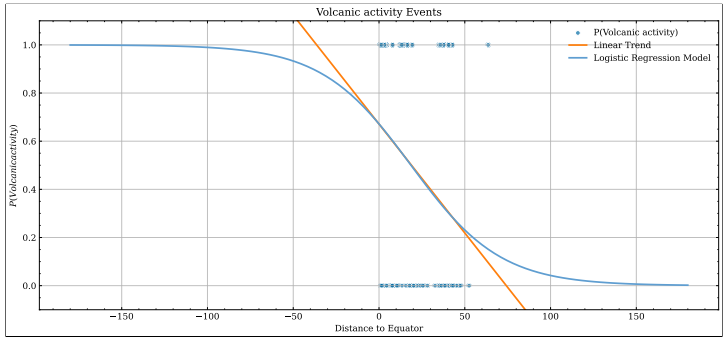


Figure 4.3: logistic fit

Logistic transformation ensures $P(X) \in [0,1]$.

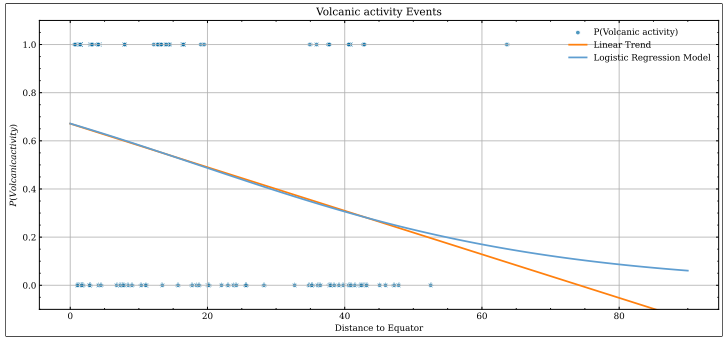


Figure 4.3: logistic fit

Agenda.

- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

Logistic regression ensures $P(X) \in [0,1]$

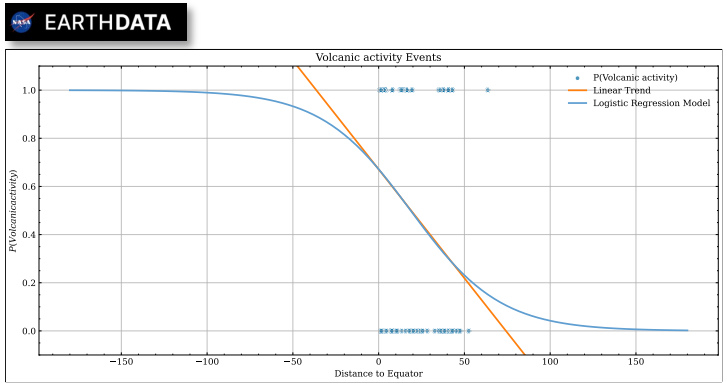


Figure 4.3: logistic fit

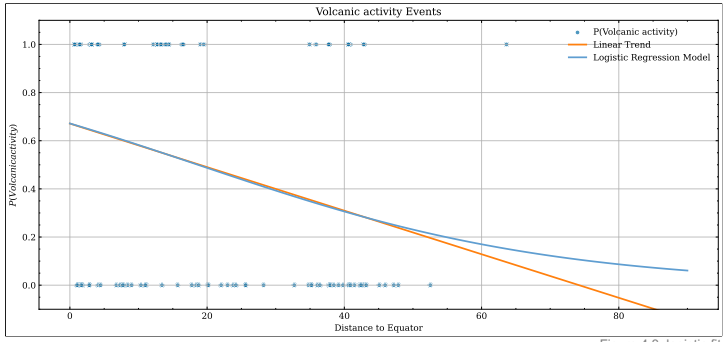


Figure 4.3: logistic fit

Let‘s write $P(X) = P(Y = 1|X)$ for short. The logistic function maps the linear predictor to a probability:

$$P(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

It's easy to see that no matter what values β_0, $\beta_1$or X take $P(X)$ will have values between 0 and 1.

A bit of rearrangement gives:

$$log\left(\frac{P(X)}{1 - P(X)}\right) = \beta_0 + \beta_1 X$$

This monotone transformation is called the log odds or logit transformation of $P(X)$.

We use maximum likelihood to estimate the parameters.

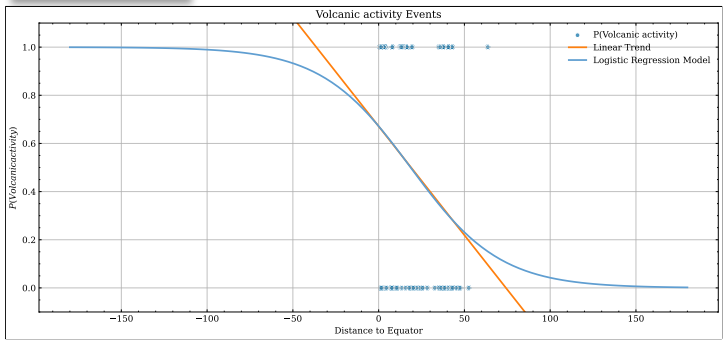


Figure 4.3: logistic fit

Logit Regression Results

Dep. Variable:	P(Volcanic activity)	No. Observations:	124
Model:	Logit	Df Residuals:	122
Method:	MLE	Df Model:	1
Date:	Fri, 17 Oct 2025	Pseudo R-squ.:	0.06352
Time:	23:17:18	Log-Likelihood:	-80.491
converged:	True	LL-Null:	-85.950
Covariance Type:	nonrobust	LLR p-value:	0.0009522

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.7163	0.290	2.474	0.013	0.149	1.284
Distance to Equator	-0.0384	0.012	-3.164	0.002	-0.062	-0.015

Logistic regression:

$$P(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}$$

We pick β_0 and β_1 to maximize the likelihood of the observed data:

$$\mathcal{L}(\beta_0,\beta_1) = \prod_{i:y_i=1} P(x_i) \times \prod_{i:y_i=0} (1 - P(x_i))$$

How do we interpret the results?

Logistic regression also works for multiple predictors and multiple target classes.

Multiple Logistic Regression

We now consider the problem of predicting a binary response using multiple predictors.

ID	Feature 1	...	Feature p	Target
1				Yes
2				No
3				No
4				Yes
5				No
...				Yes
n				Yes

By analogy to multiple linear regression, we can generalize as follows:

$$log\left(\frac{P(X)}{1-P(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

And we can rewrite to:

$$P(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

Multinomial Logistic Regression

And we can consider response variables with more than two classes:

ID	Feature 1	...	Feature p	Target
1				Flood
2				Landslide
3				Storm
4				Drought
5				Storm
...				Epidemic
n				Wildfire

$$P(Y = k|X) = \frac{e^{\beta_{0k} + \beta_{1k} X_1 + \cdots + \beta_{pk} X_p}}{\sum_{l=1}^{K} e^{\beta_{0l} + \beta_{1l} X_1 + \cdots + \beta_{pl} X_p}}$$

Feature: an individual measurable property or characteristic of a phenomenon

Target: is the variable the model is trying to predict

Logistic regression



Agenda.

- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

Discriminant analysis takes the opposite perspective on conditional distributions.

Logistic Regression directly models $P(Y = k|X)$, statistically we say we model the conditional distribution of the response Y, given the predictor(s) X.

By contrast **discriminant analysis starts with** the conditional distribution of X given Y and then uses Bayes' formula to turn things around.

$$P(Y = k|X = x)$$
$$= \frac{P(X = x|Y = k) \times P(Y = k)}{P(X = x)}$$

Why do we need discriminant analysis when we have logistic regression?

- When the classes are well-separated, the parameter estimates for the logistic regression model are surprisingly unstable. Linear discriminant analysis does not suffer from this problem.
- If n is small and the distribution of the predictors X is approximately normal in each of the classes, the linear discriminant model is again more stable than the logistic regression model.
- Linear discriminant analysis is popular when we have more than two response classes, because it also provides low-dimensional views of the data.

Recap: Lecture 2 Bayes theorem

Professorship of Business Analytics & Intelligent Systems
Chair for Information Systems and Human-centric Artificial Intelligence
Technical University of Munich, University of Bayreuth

Bayes Theorem - Motivation

OLD LEVEL OF BELIEF (Prior odds) ✖ Strength of new evidence = NEW LEVEL OF BELIEF (Posterior odds)

Posterior Probability, conditional probability of A given B

Conditional probability of B given A

Probability of A

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Probability of B

Bayes Theorem provides a way to update the probability of a hypothesis based on new evidence. The theorem considers both the initial probability of the hypothesis and how likely the new evidence is.

Define priors and density functions for discriminant analysis.

Discriminant analysis starts with the conditional distribution of X given Y and then uses Bayes' formula to turn things around.

$$P(Y = k|X = \mathrm{x}) = \frac{P(\mathrm{X} = \mathrm{x}|\mathrm{Y} = \mathrm{k}) \times P(Y = k)}{P(X = \mathrm{x})}$$

In **discriminant analysis** we write

$P(Y = k|X = \mathrm{x}) = \frac{\pi_\mathrm{k} \mathrm{f_k(x)}}{\sum_{l=1}^{K} \pi_\mathrm{l} \mathrm{f_l(x)}}$, where

$f_k(x) = P(X = x|Y = k)$ is the density for X in class k. Here we will use normal densities for these, separately in each class.

$\pi_\mathrm{k} = P(Y = k)$ is the marginal or prior probability for class k.

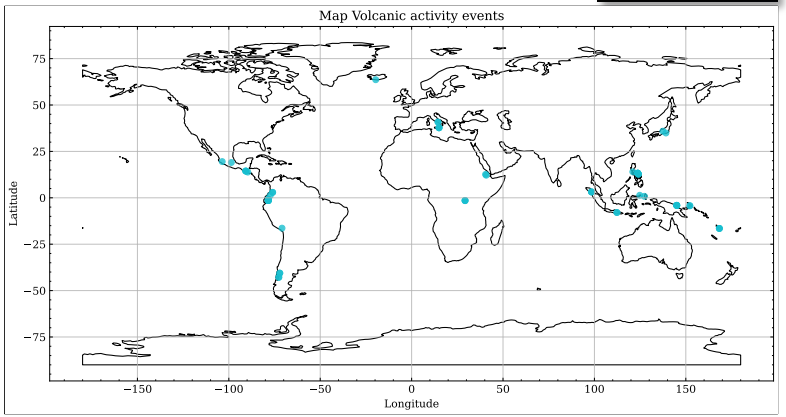


Figure 4.1: world map

ID	Distance Equator	Volcanic
1	…	Yes
2	…	No
…	…	Yes
n	13°	Yes

Total $h_{\cdot s}(f_{\cdot s})$	62 (0.03)

We assume the class densities in discriminant analysis to be Gaussian.

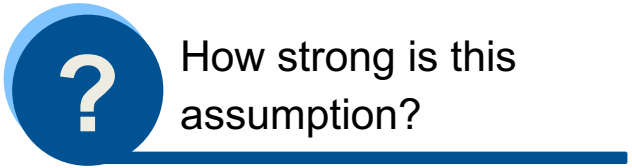
Discriminant analysis

$$P(Y = k|X = \mathrm{x}) = \frac{\pi_\mathrm{k} \times \mathrm{f_k(x)}}{\sum_{l=1}^{K} \pi_\mathrm{l} \mathrm{f_l(x)}}$$

The **gaussian density** function has the form:

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_\mathrm{k}} \mathrm{e}^{-\frac{1}{2}\left(\frac{\mathrm{x}-\mu_\mathrm{k}}{\sigma_\mathrm{k}}\right)^2}$$, where

μ_k is the mean and ${\sigma_\mathrm{k}}^2$ the variance (in class k).
We will assume that all $\sigma_\mathrm{k} = \sigma$ are the same.



EARTHDATA

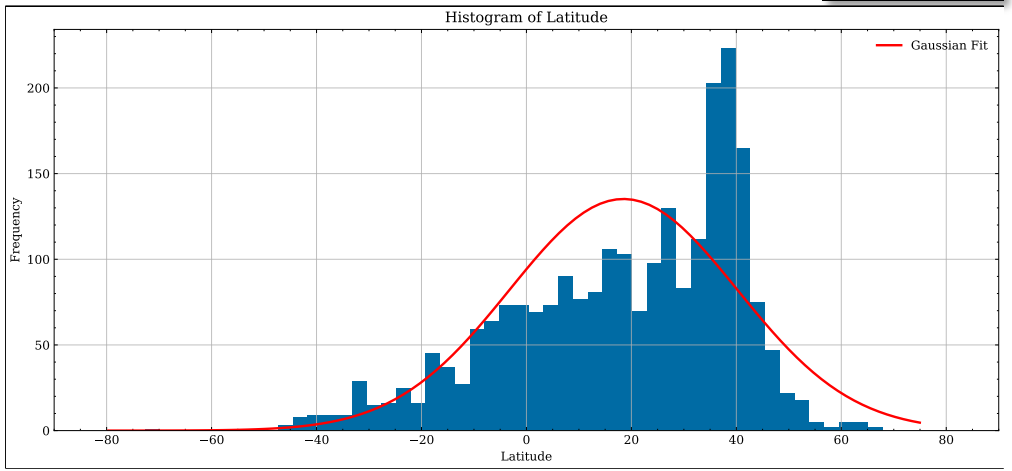


Figure 4.4: class prior

We get:

$$P(Y = k|X = \mathrm{x}) = \frac{\pi_\mathrm{k} \times \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{1}{2}\left(\frac{\mathrm{x}-\mu_\mathrm{k}}{\sigma}\right)^2}}{\sum_{l=1}^{K} \pi_\mathrm{l} \times \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{1}{2}\left(\frac{\mathrm{x}-\mu_\mathrm{l}}{\sigma}\right)^2}}$$

We can break down how to estimate parameters and run inference.

Discriminant analysis

$$P(Y = k|X = \mathrm{x}) = \frac{\pi_\mathrm{k} \times \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{1}{2}\left(\frac{\mathrm{x}-\mu_\mathrm{k}}{\sigma}\right)^2}}{\sum_{l=1}^{K} \pi_\mathrm{l} \times \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{1}{2}\left(\frac{\mathrm{x}-\mu_\mathrm{l}}{\sigma}\right)^2}}$$

We **estimate parameters** as follows:

$$\hat{\pi}_\mathrm{k} = \frac{\mathrm{n_k}}{\mathrm{n}}$$

$$\hat{\mu}_\mathrm{k} = \frac{1}{\mathrm{n_k}} \sum_{\mathrm{i:\, y_i = k}} \mathrm{x_i}$$

$$\hat{\sigma}^2 = \frac{1}{\mathrm{n} - \mathrm{K}} \sum_{\mathrm{k=1}}^{\mathrm{K}} \sum_{\mathrm{i:\, y_i = k}} (\mathrm{x_i} - \hat{\mu}_\mathrm{k})^2$$

Running inference

To classify $X = x$, we want to know k for which $P(Y = k|X = \mathrm{x})$ is largest.

Taking logs and discarding terms that do not depend on k we get the linear function of x:

$$\delta_k(x) = x \times \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + log(\pi_k)$$



What changes if X is not just a one-dimensional predictor?

LDA



Agenda.

- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

What changes if X is not just a one-dimensional predictor?

Multivariate Gaussian distribution

We now extend the linear discriminant analysis to multiple predictors. We assume that $X = (X_1, \dots, X_p)$ is drawn from a multivariate gaussian distribution.

We write:

$$X \sim N(\mu, \Sigma)$$

With μ being the mean of X, a p-dimensional vector and Σ the $p \times p$ covariance matrix.

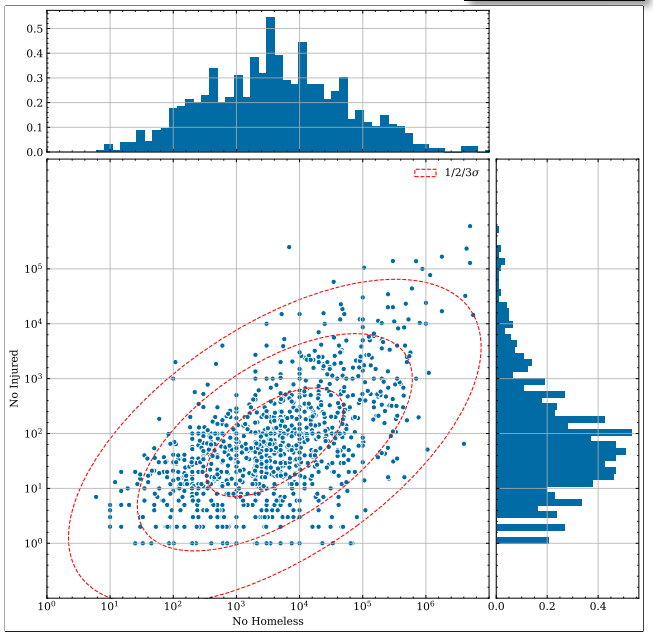


Figure 4.5: multivariate

So, what changes?

$$P(Y = k | X = \mathrm{x}) = \frac{\pi_k \times \mathrm{f_k(x)}}{\sum_{l=1}^{K} \pi_l \mathrm{f_l(x)}}$$

$$X \sim N\left(\mu = \begin{matrix} 3.61 \\ 1.84 \end{matrix}, \Sigma = \begin{matrix} 1.19 & 0.63 \\ 0.63 & 0.98 \end{matrix}\right)$$

What changes if X is not just a one-dimensional predictor?

Multivariate Gaussian distribution

$$P(Y = k|X = \mathrm{x}) = \frac{\pi_\mathrm{k} \times \mathrm{f_k(x)}}{\sum_{l=1}^{K} \pi_\mathrm{l} \mathrm{f_l(x)}}$$

The **gaussian density** function now has the form:

$$f_k(x) = \frac{1}{(2\pi)^{\frac{\mathrm{p}}{2}} |\Sigma|^{\frac{1}{2}}} \mathrm{e}^{-\frac{1}{2}(\mathrm{x}-\mu_k)^\mathrm{T}\Sigma^{-1}(\mathrm{x}-\mu_k)}$$

And despite its complexity the **discriminant function** still has a linear form:

$$\delta_k(x) = \mathrm{x}^\mathrm{T}\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\mathrm{T}\Sigma^{-1}\mu_k + log(\pi_k)$$

EARTHDATA

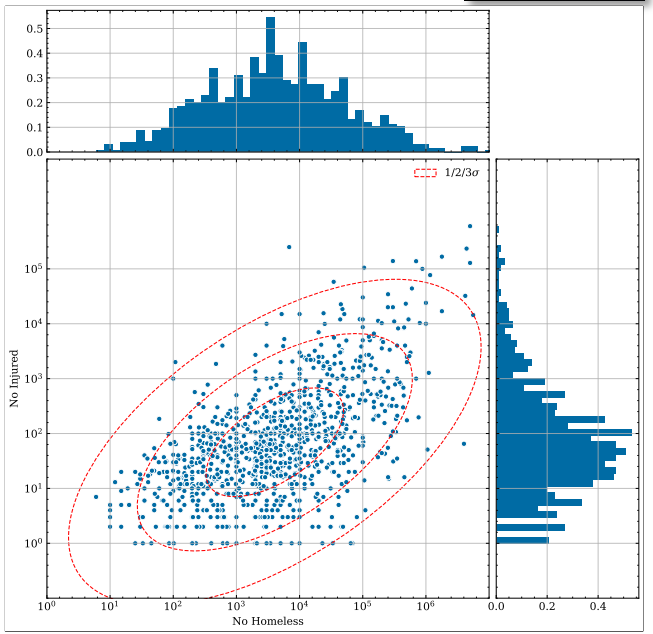


Figure 4.5: multivariate

$$X \sim N\left(\mu = \begin{matrix} 3.61 \\ 1.84 \end{matrix}, \Sigma = \begin{matrix} 1.19 & 0.63 \\ 0.63 & 0.98 \end{matrix}\right)$$

Different forms of Discriminant Analysis based on our assumptions.

Discriminant analysis

$$P(Y = k|X = \mathrm{x}) = \frac{\pi_k \mathrm{f_k(x)}}{\sum_{l=1}^{K} \pi_l \mathrm{f_l(x)}}$$

$\pi_k = P(Y = k)$ is the marginal or prior probability for class k. → its computation is straightforward.

$f_k(x) = P(X = x|Y = k)$ is the density for X in class k → that is the challenging part, where we discussed various assumptions.

Our assumptions:	What we need to estimate in $f_k(x)$:
Without any assumptions, **Bayes' theorem**.	K p-dimensional density functions.
If we assume $f_k(x)$ to be gaussian and Σ to be the same covariance matrix for each class, this leads to **linear discriminant analysis**.	K p-dimensional mean vectors and one $p \times p$-dimensional covariance matrix.
With gaussians but different Σ_k in each class, we get **quadratic discriminant analysis**.	K p-dimensional mean vectors and K $p \times p$-dimensional covariance matrices.
With $f_k(x) = \prod_{j=1}^{p} f_{jk}(\mathrm{x_j})$, conditional independence in each class, we get **naïve Bayes**. For Gaussian this means Σ_k are diagonal.	? What changes?

QDA



Agenda.

- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

Why do we want to make the independence assumption for Naïve Bayes?

Discriminant analysis

$$P(Y = k|X = \mathrm{x}) = \frac{\pi_k f_k(\mathrm{x})}{\sum_{l=1}^{K} \pi_l f_l(\mathrm{x})}$$

With $f_k(x) = \prod_{j=1}^{p} f_{jk}(x_j)$, conditional independence in each class, we get **naïve Bayes**. For Gaussian this means Σ_k are diagonal.

What we need to estimate in $f_k(x)$:

K p-dimensional mean vectors and K p-dimensional variance vectors.

Motivation

In the case of a multivariate normal distribution, the association between the different predictors is summarized by the off-diagonal elements of the covariance matrix.

This association can be very hard to characterize and exceedingly challenging to estimate. If we want to compute the conditional probability $P(Y = k|X = \mathrm{x})$ for every observation $X = (X_1, \ldots, X_p)$, we may not be able to observe every combination of observations and target classes. When p is large multivariate methods like **QDA** and even **LDA** break down.

By assuming that the p covariates are independent within each class, we completely eliminate the need to worry about the association between the p predictors.

Naïve Bayes – Conditional Independence

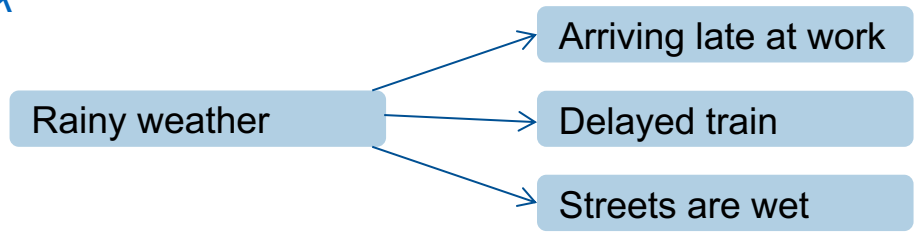
Probabilities of some events are not independent, e.g., the probability of a *delayed train (D)* and *arriving late at work (A)* are not independent (but positively correlated)

Probability that both D and A happen → $\mathrm{P}(\mathrm{D} \cap \mathrm{A}) > \mathrm{P}(\mathrm{D})\,\mathrm{P}(\mathrm{A})$

However, these events may be **conditionally independent** if we have additional knowledge about the situation, e.g., that there is *rainy weather*

$$\mathrm{P}(\mathrm{D} \cap \mathrm{A}|\mathrm{R}) = \mathrm{P}(\mathrm{D}|\mathrm{R})\,\mathrm{P}(\mathrm{A}|\mathrm{R})$$

In this setting, the *rainy weather* explains the dependence between *delayed train* and *arriving late at work*



Naïve Bayes – Continuous Example

Task: Distinguish cats from dogs based on their size

- Classes cats and dogs: $Y \in \{c, d\}$
- Features height [cm], weight [kg]: $\mathrm{X_i} = (\mathrm{h_i}, \mathrm{w_i})$
- Training examples: $(\mathrm{X_i}, \mathrm{Y_i}) = (\mathrm{h_i}, \mathrm{w_i}, \mathrm{y_i})$ with 4 dogs and 12 cats



[1]

Given probabilities: $\mathrm{P(d)} = \frac{4}{4+12} = 0.25$, $\mathrm{P(c)} = \frac{12}{4+12} = 0.75$

Model for dogs:

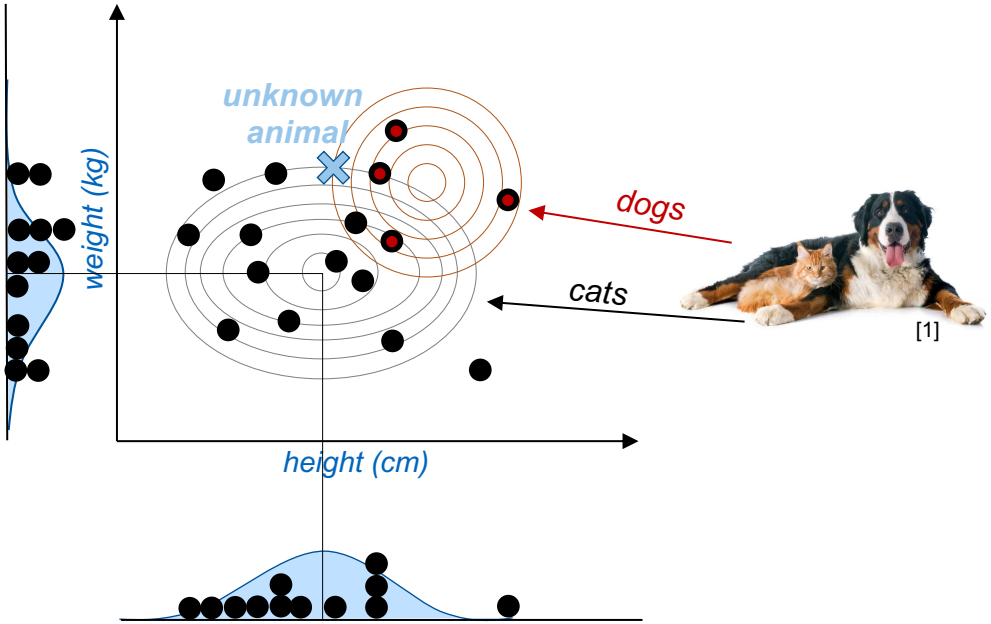
- Height ~ Gaussian with mean and variance
- Weight ~ Gaussian $(\hat{\mu}_{\mathrm{w,d}}\ , \hat{\sigma}_{\mathrm{w,d}}{}^2)$
- Assume that height and weight are independent

$$\begin{cases} \hat{\mu}_{\mathrm{h,d}} = \frac{1}{4} \sum_{\mathrm{i:y_i=d}} \mathrm{h_i} \\ \hat{\sigma}_{\mathrm{h,d}}{}^2 = \frac{1}{3} \sum_{\mathrm{i:y_i=d}} (\mathrm{h_i} - \hat{\mu}_{\mathrm{h,d}})^2 \end{cases}$$

Model for cats:

- Same, using $(\hat{\mu}_{\mathrm{h,c}}, \hat{\sigma}_{\mathrm{h,c}}{}^2)$ and $(\hat{\mu}_{\mathrm{w,c}}, \hat{\sigma}_{\mathrm{w,c}}{}^2)$

Naïve Bayes – Continuous Example cont‘d



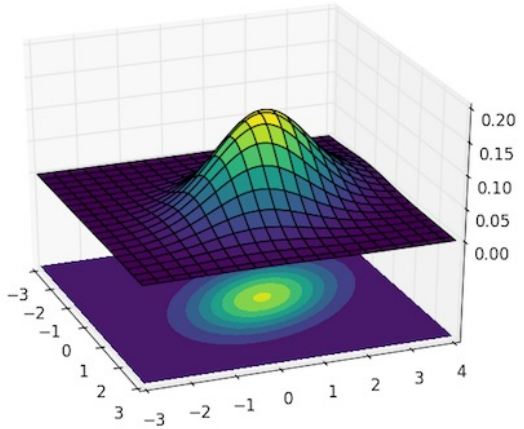
Recap: Gaussian Distribution

Probability density function:

distribution mean

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}$$

standard deviation



[2]

Discriminant analysis starts with one predictor and the class densities.

Bayes' theorem for classification

$$P(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

$\pi_k = P(Y = k)$ is the prior of class k, $f_k(x) = P(X = x|Y = k)$ its density.

Setting

One predictor x (here log $_{10}$ Total Affected), $K = 2$ classes: volcanic activity or other.

We only look at the empirical distribution of x in each class.

Assumption on this slide

None on $f_k(x)$. Without a model for the class densities we cannot evaluate Bayes' theorem.

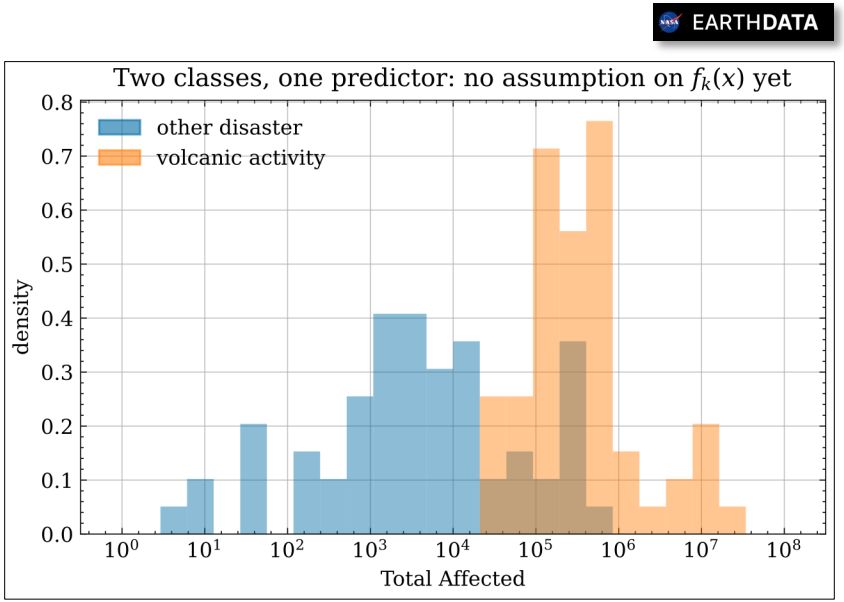


Figure 4.6: discriminant assumptions

LDA assumes Gaussian class densities with one variance shared by all classes.

Assumption on this slide (LDA)

$f_k(x) = \mathcal{N}(\mu_k, \sigma^2), \qquad \sigma_k = \sigma \text{ for all } k$

Class density and estimates

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma}\right)^2}$$

with $\hat{\mu}_k = \frac{1}{n_k}\sum_{i:y_i=k} x_i$ and $\hat{\sigma}^2 = \frac{1}{n-K}\sum_{k=1}^{K}\sum_{i:y_i=k}(x_i - \hat{\mu}_k)^2$

Discriminant function

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{{\mu_k}^2}{2\sigma^2} + \log(\pi_k)$$

Linear in x: the term in x^2 is the same for every class and drops out.

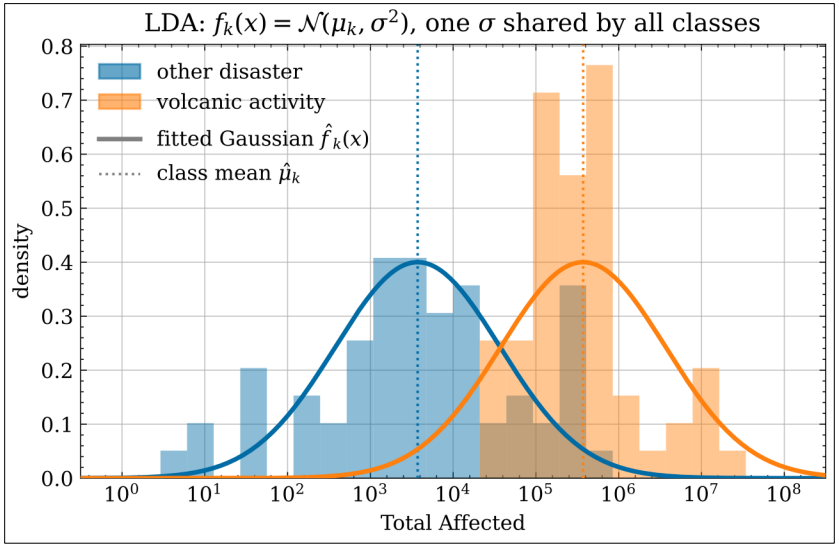


Figure 4.6: discriminant assumptions

QDA keeps a separate variance per class, and the discriminant becomes quadratic.

Assumption on this slide (QDA)

$f_k(x) = \mathcal{N}(\mu_k, \sigma_k^2)$

Class density and estimates

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \mathrm{e}^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}$$

with $\hat{\mu}_k = \frac{1}{n_k}\sum_{i:y_i=k} x_i$ and $\hat{\sigma}_k^2 = \frac{1}{n_k - 1}\sum_{i:y_i=k}(x_i - \hat{\mu}_k)^2$

Discriminant function

$$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + x\frac{\mu_k}{\sigma_k^2} - \frac{{\mu_k}^2}{2\sigma_k^2} - \log(\sigma_k) + \log(\pi_k)$$

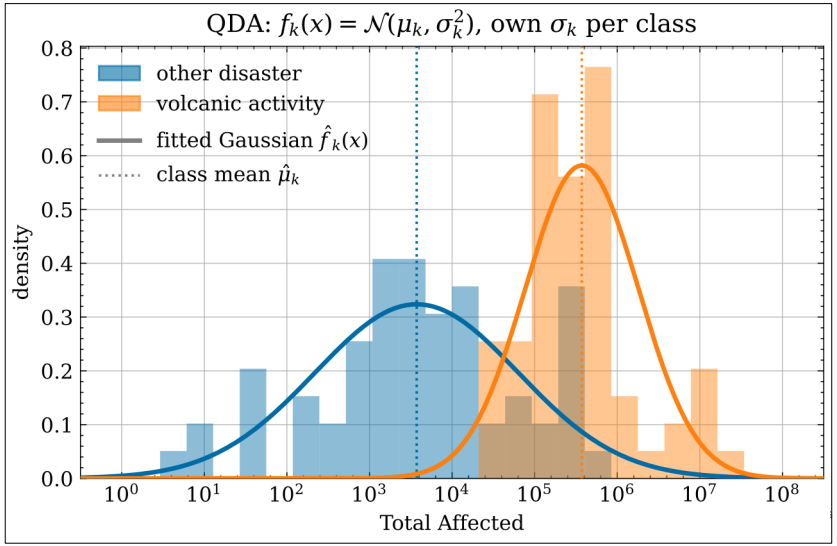


Figure 4.6: discriminant assumptions

With two predictors, each class density lives in the plane.

Setting

$x = (x_1, x_2)^{\mathrm{T}}, \qquad p = 2$

The 1-D predictor becomes the second coordinate; log $_{10}$ Total Deaths is added as the first.

What changes

The class density is now a function of two arguments, $f_k(x_1, x_2)$. Its shape is what the three methods assume differently.

Assumption on this slide

None on $f_k(x)$. The scatter shows the empirical class distributions.

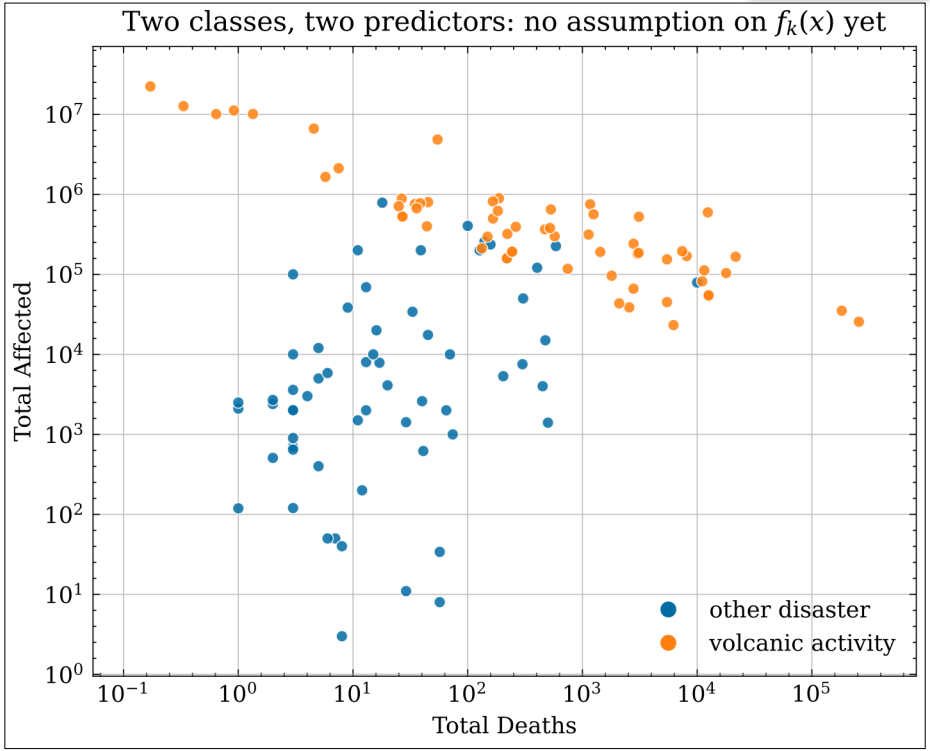


Figure 4.6: discriminant assumptions

LDA with p predictors: one covariance matrix shared by all classes.

Assumption on this slide (LDA)

$f_k(x) = \mathcal{N}(\mu_k, \Sigma), \qquad \Sigma_k = \Sigma \text{ for all } k$

Class density

$$f_k(x) = \frac{1}{(2\pi)^{\frac{p}{2}}|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k)}$$

Σ is the $p \times p$ covariance matrix; off-diagonal entries may be non-zero, so the contours are tilted ellipses.

Discriminant function

$$\delta_k(x) = x^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k{}^T\Sigma^{-1}\mu_k + \log(\pi_k)$$

Still linear in x. We estimate K mean vectors and one Σ: $Kp + p(p+1)/2$ numbers.

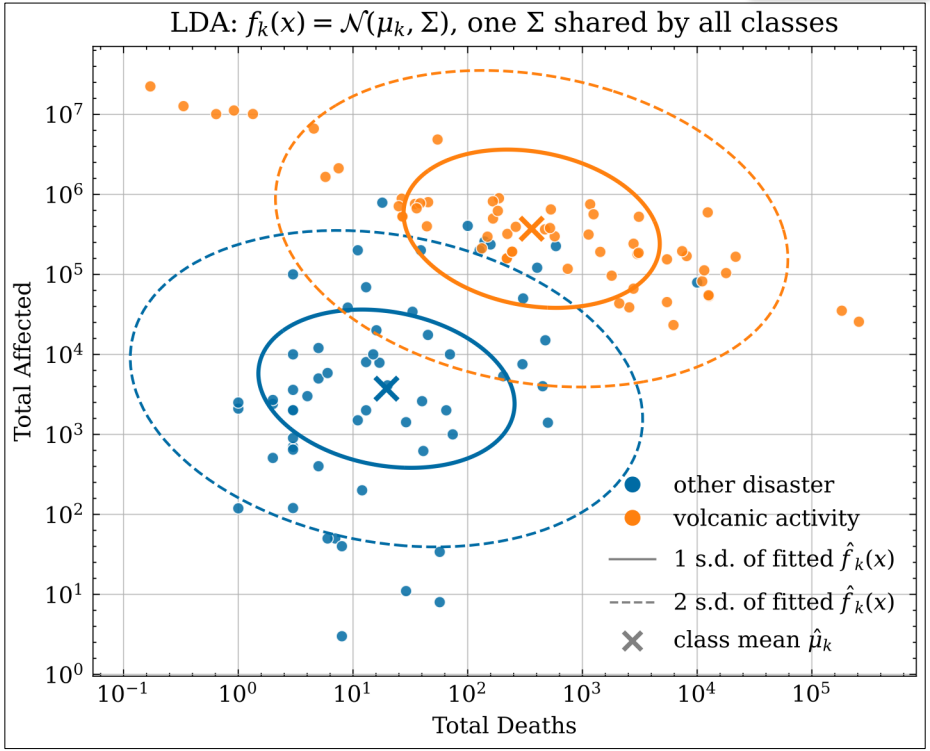


Figure 4.6: discriminant assumptions

QDA with p predictors: a separate covariance matrix for each class.

Assumption on this slide (QDA)

$f_k(x) = \mathcal{N}(\mu_k, \Sigma_k)$

Discriminant function

$\delta_k(x)$

$= -\frac{1}{2} x^T {\Sigma_k}^{-1} x + x^T {\Sigma_k}^{-1} \mu_k - \frac{1}{2} \mu_k^T {\Sigma_k}^{-1} \mu_k - \frac{1}{2} \log|\Sigma_k|$

$+ \log(\pi_k)$

Quadratic in x: the orange term differs between classes. We estimate K mean vectors and K matrices: $Kp + Kp(p+1)/2$ numbers.

What the figure shows

Each ellipse has its own size and tilt. QDA is more flexible than LDA but needs more data per class.

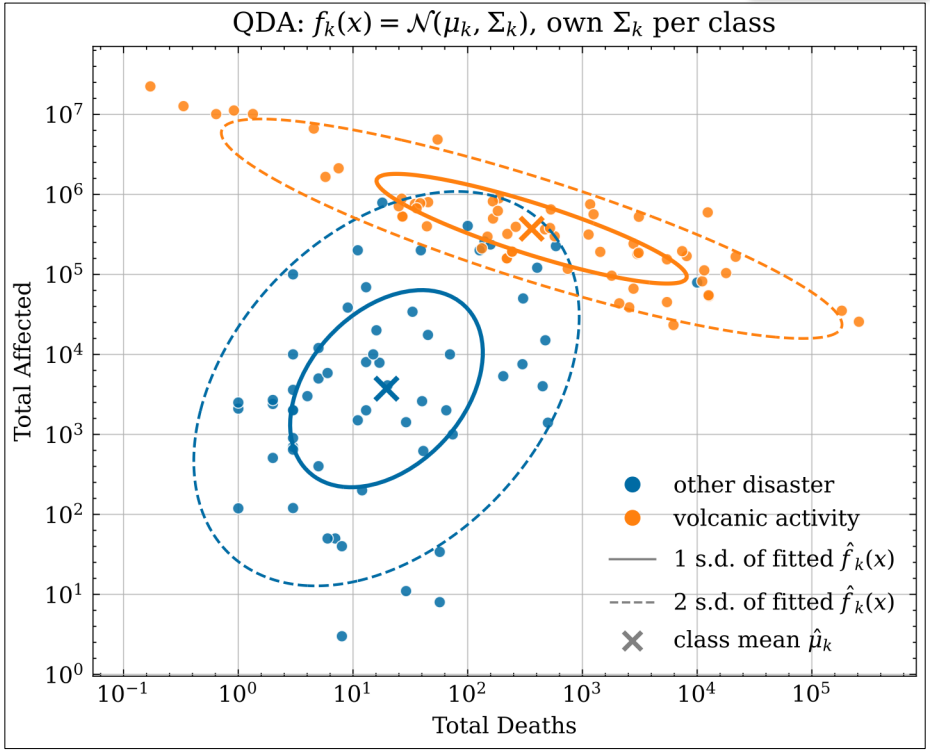


Figure 4.6: discriminant assumptions

Naïve Bayes assumes the predictors to be independent within each class.

Assumption on this slide (naïve Bayes)

$$f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$$

Class density

With Gaussian factors this is QDA with a diagonal covariance matrix:

$\Sigma_k = \mathrm{diag}(\sigma_{k1}^2, \ldots, \sigma_{kp}^2)$

The off-diagonal entries are set to zero, so the contours are ellipses aligned with the axes.

Discriminant function

$$\delta_k(x) = \sum_{j=1}^{p} \left[-\frac{(x_j - \mu_{kj})^2}{2\sigma_{kj}^2} - \log\sigma_{kj} \right] + \log(\pi_k)$$

Quadratic in each x_j separately. We estimate $2Kp$ numbers, which grows linearly in p.

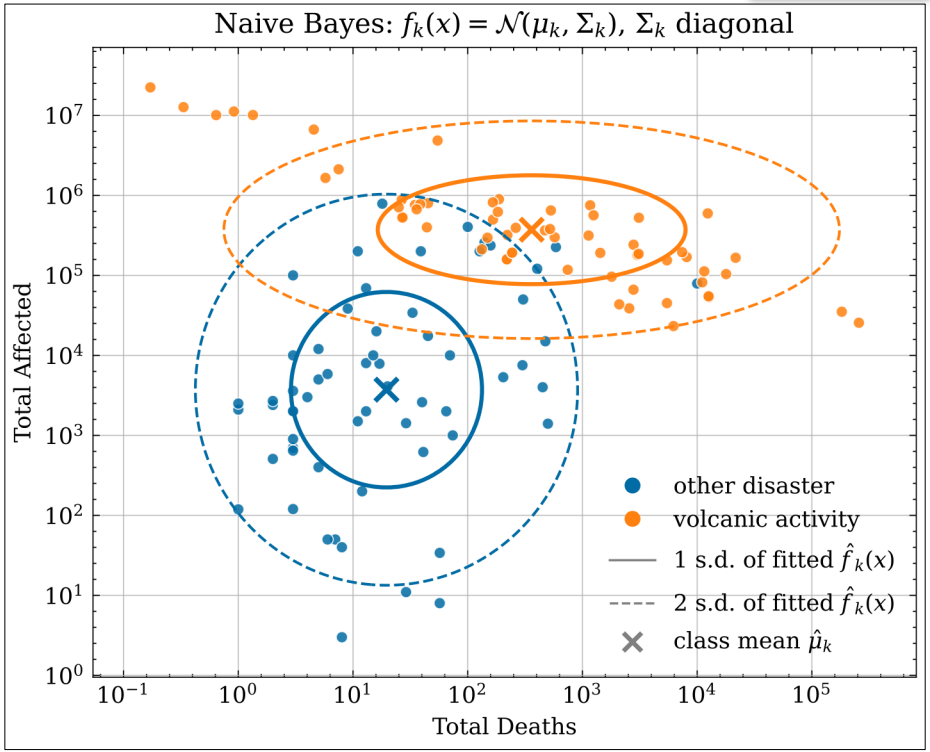


Figure 4.6: discriminant assumptions

Naïve Bayes

